Team TYS

FinTech / AI in Finance · final defence

Mode: Partner product — MKBANK, micro & family-business credit department

Partners: MKBANK and the Central Bank of Uzbekistan · October 2026

Live prototype https://tys-fintech.vercel.app **Repository** https://github.com/hmunisbekk-droid/TYS-fintech

The regulation names the evidence — the self-employed have exactly one line

CBU regulation on the debt-burden ratio, cl.9 — income a lender may rely on

- **Employer-confirmed salary** — no employer — the borrower is the business
- **Pension** — working-age applicant, not a pensioner
- **Funded-pension contributions** — none: contributions come through an employer
- ✓ **Income credited to the borrower's bank accounts** — the only line a self-employed borrower has
- **Taxes paid** — self-employed income is untaxed; no return is filed
- **Interest, dividends, rental income** — rarely applicable; easy to document when it is

50% cap

ПДН limit for all loans since 1 Jan 2025 (cl.10); 12-month arithmetic mean of income (cl.7)

~140,000

micro-projects MKBANK targets in 2026 (47,975 in 2025); 1,062 mahalla agents

60–90 min

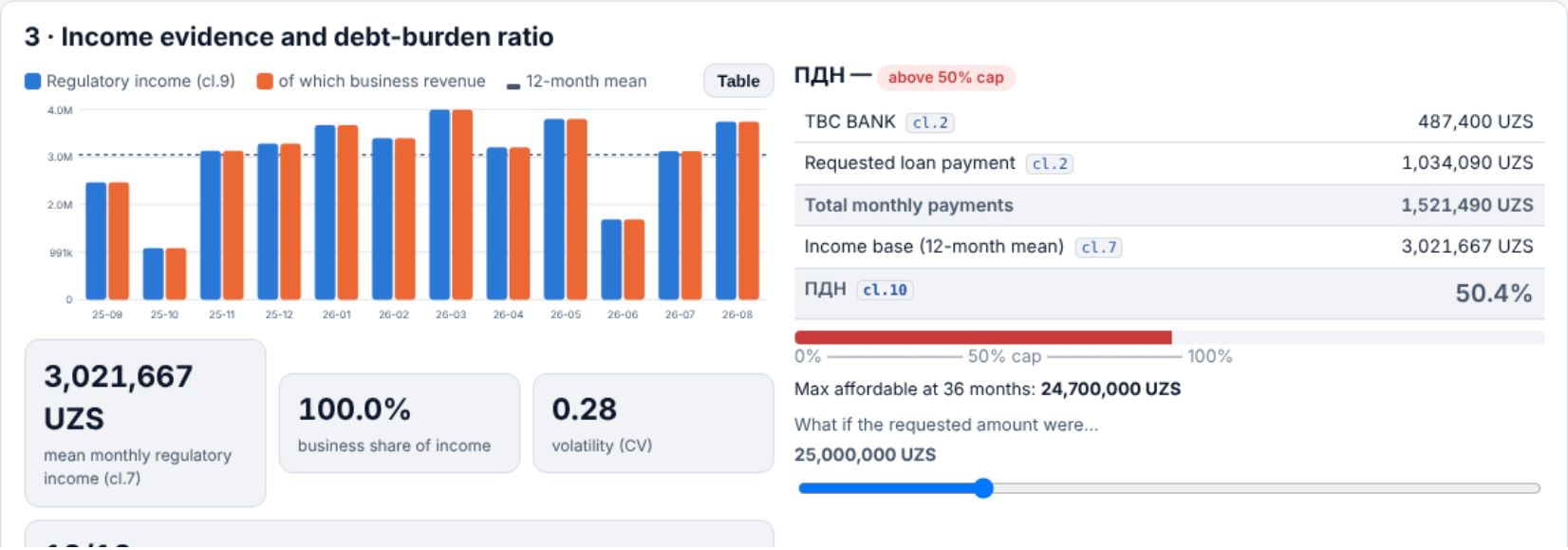
an officer's manual reading of 12 months of statements per application (our estimate; pilot measures it)

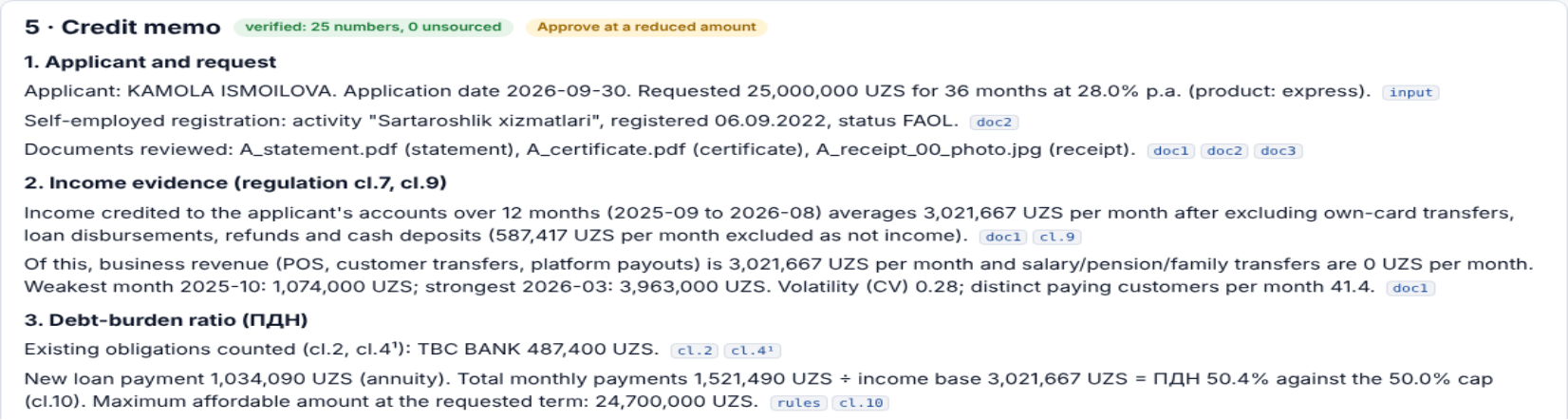
0 lines

of official income records for ~790k registered self-employed (untaxed, unfiled, no cash register required)

Sources: lex.uz/docs/4654149 and 6809992 (CBU regulation and 2024 amendments) · spot.uz 27 Mar 2026 (MKBANK, bank's own figures) · yuz.uz (self-employed registrations, 2021) · buxgalter.uz

What the officer gets in under five seconds





12-month regulatory income

cl.7 mean, cl.9 definition; own-card transfers, loan proceeds, refunds and cash deposits excluded

ПДН with citations

36-month spreading, guarantor rule, family-programme exemption, lender-type rules (banks: no small-loan exclusion, 15% quota is portfolio-level), 50% cap, maximum affordable amount

Integrity report

balance chain, duplicates, coverage, gaps, pre-application spikes, circular transfers, holder match

Verified memo

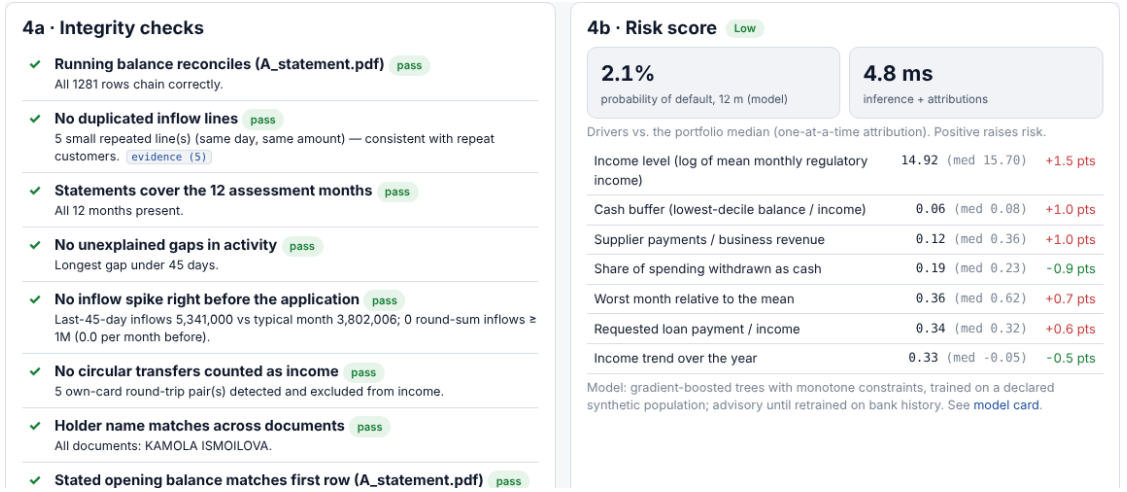
every number traces to its source — statement lines (click to open them), a clause, or the model; the verifier refuses unsourced numbers; hash-chained audit log

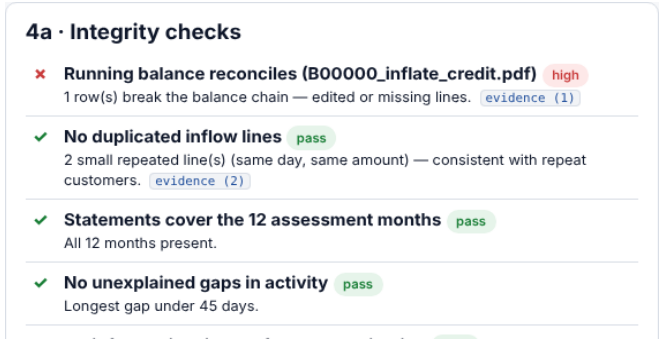
Live demo — on our samples and on anything you upload

Open on your laptop or phone

https://tys-fintech.vercel.app

1. Load sample A — statement PDF, Soliq certificate, receipt photo parse in the tab (no server, no API)
2. Income series, ПДН with clause citations, maximum affordable amount; drag the slider to cross the 50% cap
3. Integrity checks and risk score with drivers; click any evidence link to see the statement line
4. Memo: verifier badge shows how many numbers were checked and that none is unsourced
5. Record the decision; export the case with its hash-chained audit log
6. Then: Load tampered statement — the balance chain breaks at the inflated inflow; upload your own documents





Scan to open the live prototype on your phone (New case → Load sample A)

Seven components, one optional language model, zero API keys

1 · Intake

pdf.js with coordinates; XLSX/CSV; tesseract.js OCR uzb+rus+eng, self-hosted; field confidence, tap-to-confirm

2 · Categoriser

multilingual rules + counterparty patterns (recurring payer → family; same mask both ways → own card)

3 · Cash-flow

12-month series; 18 features identical in TypeScript and Python (parity test)

4 · Rule engine

ПДН clause by clause: 36/180-month spreading, guarantor rule, lender-type exclusions, family-programme exemption, 50% cap, back-solved amount

5 · Risk model

XGBoost, monotone constraints, JSON trees evaluated in the browser; attributions vs portfolio median

6 · Integrity

balance chain, duplicates, coverage, gaps, round-sum spikes, circular transfers, holder match

7 · Memo + audit

template from a fact table; verifier rejects unsourced numbers; SHA-256 hash-chained log

Perimeter

Prototype: static site, every byte of a document stays in the browser. Production: the same bundle on the bank's server, SSO, audit log in the bank DB, core-banking pull for own customers. LLM optional, narrative only, behind the same verifier.

Speed: statement 1.2–1.8 s · receipt OCR 1.6 s · everything else < 50 ms · marginal cost per application: zero

Human in the loop: the engine recommends and computes the maximum affordable amount; the officer decides; an override needs a reason, written into the chain.

Measured through the real app, not typed into a slide

60/60

generator PDFs: exact rows, balance chain reconciled (2 layouts, RU/UZ)

8/8

unseen layouts exact — CSV/XLSX/TSV variants incl. a wallet export (list on the model card)

40/40

tampered statements caught, 4 tamper types; 0 false alarms on clean ones

95%

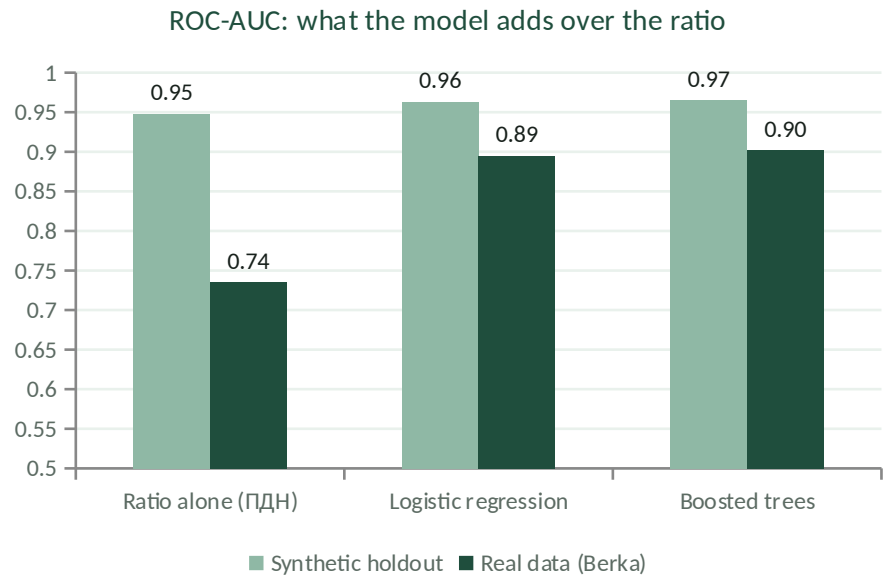
receipt OCR total exact, 60 camera-augmented photos; INN 100%, date 97%

0.965

ROC-AUC, synthetic holdout n=500; logistic 0.963, ratio alone 0.948

0.902

ROC-AUC, real bank data (Berka, 443 loans, 44 bad; 95% CI ≈ 0.86–0.94); ratio alone 0.735



Honest reading: on synthetic data the ratio already carries most of the signal by construction; on real statements the cash-flow features add +0.17 AUC over the ratio, and logistic regression comes close to the trees — we ship the trees for the monotone constraints and non-linear thresholds, not for the last 0.01. Categoriser: 99.96% on 10,481 lines of the generator's own vocabulary — will fall on real data. Memo verifier: 0 unsourced numbers across a 300-scenario sweep (amounts × terms × products × borrowers), enforced at run time.



All numbers are read from JSON produced by ml/train.py, ml/berka_validate.py, npm test and tests/*_eval.py (see the Model card page).

Caveat on the slide, not in the footnote: the risk score is advisory until retrained on the bank's own history inside the perimeter — that is the pilot's first deliverable.

No partner data, so we built a declared generator

1

Archetypes

10 trades (tailor, taxi, dehkan farmer, grocer, reseller, tutor…) with declared seasonality, customer counts, channel mix, expense structure

2

Simulation

2,000 borrowers · 5,689,589 transactions · existing loans with annuity schedules · shocks · default label from a latent affordability score (10.5% rate)

3

Rendering

60 statement sets as PDF (text layer) + XLSX + CSV in RU and UZ layouts; 180 fiscal receipts; Soliq-style certificates with QR

4

Camera model

perspective, rotation, lighting gradient, blur, sensor noise, JPEG — then measured through the same OCR the officer uses

5

Tampering

40 statements: inflated inflow, deleted debit, duplicated row, window dressing — labelled for the integrity evaluation



HUMO CARD STATEMENT
Выписка по карте

Владелец: KAMOLA ISMOILOV
Карта: 5614 **** **** 4369
Период: 2025-09-01 — 2026-08-31
Валюта: UZS

Входящий остаток: **668 582.52**

Дата	Время	Описание операции	Расход	Приход	Остаток
2025-09-01	07:33	P2P HUMO 8600 **** 9613		50 000.00	718 582.52
2025-09-01	08:33	IJARA TO'LOVI P2P 8600 **** 2210	674 000.00		44 582.52
2025-09-01	08:54	NAQD PUL KIRITISH BANKOMAT		350 000.00	394 582.52
2025-09-01	09:03	HUDUDGAZTAMINOT KOMMUNAL	131 000.00		263 582.52
2025-09-01	14:57	HUDUDIY ELEKTR TARMOQLARI KOMMUNAL	90 000.00		173 582.52
2025-09-01	16:24	UCELL ABONENT	62 000.00		111 582.52
2025-09-02	14:09	BARAKA MARKET TOSHKENT	103 000.00		8 582.52
2025-09-03	16:36	POS KAMOLA SARTAROSHLIK TERMINAL		52 000.00	60 582.52
2025-09-04	07:29	P2P HUMO 9860 **** 9719		17 000.00	77 582.52
2025-09-04	12:45	POS KAMOLA SARTAROSHLIK TERMINAL		40 000.00	117 582.52
2025-09-04	13:46	POS KAMOLA SARTAROSHLIK TERMINAL		22 000.00	139 582.52
2025-09-04	16:24	P2P HUMO 8600 **** 6177		38 000.00	177 582.52
2025-09-04	16:48	P2P HUMO 9860 **** 4617		39 000.00	216 582.52
2025-09-04	17:40	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	243 582.52
2025-09-04	22:21	ANGLESEY TOSHKENT	63 000.00		180 582.52
2025-09-05	20:06	POS KAMOLA SARTAROSHLIK TERMINAL		32 000.00	212 582.52
2025-09-05	20:29	P2P HUMO 8600 **** 9240		31 000.00	243 582.52
2025-09-05	22:57	OPTOM SAVDO MCHJ TO'LOV	104 000.00		139 582.52
2025-09-06	14:53	POS KAMOLA SARTAROSHLIK TERMINAL		64 000.00	203 582.52
2025-09-06	17:24	HAVAS TOSHKENT	131 000.00		72 582.52
2025-09-07	11:40	P2P HUMO 9860 **** 3683		66 000.00	138 582.52
2025-09-07	13:08	POS KAMOLA SARTAROSHLIK TERMINAL		31 000.00	169 582.52
2025-09-07	20:29	OPTOM SAVDO MCHJ TO'LOV	88 000.00		81 582.52
2025-09-08	07:44	POS KAMOLA SARTAROSHLIK TERMINAL		43 000.00	124 582.52
2025-09-08	15:53	POS KAMOLA SARTAROSHLIK TERMINAL		39 000.00	163 582.52
2025-09-08	16:34	OPTOM SAVDO MCHJ TO'LOV	106 000.00		57 582.52
2025-09-08	18:41	KORZINKA TOSHKENT	42 000.00		15 582.52
2025-09-09	08:34	POS KAMOLA SARTAROSHLIK TERMINAL		50 000.00	65 582.52
2025-09-09	09:50	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	92 582.52
2025-09-09	21:57	HAVAS TOSHKENT	50 000.00		42 582.52
2025-09-10	12:30	POS KAMOLA SARTAROSHLIK TERMINAL		30 000.00	72 582.52
2025-09-10	18:43	POS KAMOLA SARTAROSHLIK TERMINAL		46 000.00	118 582.52
2025-09-10	22:08	P2P HUMO 8600 **** 3739		34 000.00	152 582.52
2025-09-11	10:18	P2P HUMO 9860 **** 5631		33 000.00	185 582.52
2025-09-11	17:02	POS KAMOLA SARTAROSHLIK TERMINAL		37 000.00	222 582.52
2025-09-11	18:42	P2P HUMO 8600 **** 7135		44 000.00	266 582.52
2025-09-11	21:26	P2P HUMO 9860 **** 6338		30 000.00	296 582.52
2025-09-11	22:12	P2P HUMO 9860 **** 6674		34 000.00	330 582.52
2025-09-12	15:07	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	357 582.52
2025-09-12	18:20	P2P HUMO 9860 **** 2803		24 000.00	381 582.52



KAMOLA SARTAROSHLIK
Xorazm, Mustaqillik ko'chasi 9
STIR/INN: 342056057
FISKAL CHEK № 58332
01.03.2026 13:36
Kamola
Sana
Kassir
BOSH YUVISH 45 000
3 x 15 000
SOCH OLISH 40 000
1 x 40 000
JAMI 85 000
To'landi (UZCARD) 85 000
QQS 12% 9 107
FM: UZ118914450618 FP: 135878439006
ofd.soliq.uz
Xaridingiz uchun rahmat!

Synthetic statement (Humo-style layout, Russian labels) and a camera-augmented synthetic fiscal receipt. No real person is represented; names come from a short public list, card numbers are random and masked.

```
Regenerate:  python -m synth.cli --n 2000 --render 60 --
receipts 180 --tamper 40 --seed 42
```

Also: methodology check on real anonymised bank data (Berka, PKDD'99) — not redistributed, downloaded by script.

Pilot: three months, five branches, fifty mahalla agents

Cost to build and run the pilot (assumptions, UZS)

Line	Assumption	3-month cost
Team: 3 engineers/analysts (TYS)	3 × 3 months × 15M	135M
Bank credit analyst, part-time	0.5 FTE × 3 months × 15M	22.5M
Infrastructure	1 VM in the bank DC, no GPU	≈ 3M
Integration (core banking, bureau)	bank IT, 20 person-days	≈ 30M
Data, licences, API spend	none — open-source, no external API	0
Total pilot (≈ $15k)		**≈ 190M**

Running cost per application: zero marginal (static bundle; no API, no GPU). Optional narrative LLM on one on-prem GPU: well under 1 US cent per application.

Timeline: month 1 shadow mode (engine runs beside the manual process; we measure minutes and agreement) · months 2–3 assisted mode (officer works from the memo) · success metrics: minutes per application, agreement with committee, override rate, ПДН audit findings, retrained model AUC on bank history.

Value — the officer-hours formula (every term is a slider)

90 → 25 min

document handling per application, before → with Daromad (assumption 1; the pilot measures it)

≈ 42,000 / yr

addressable applications: 30% of MKBANK's 140,000 micro-projects (bank's own 2026 target) — those outside the ПДН-exempt family programme (assumption 2)

≈ 45,000 h

officer-hours saved per year = 42,000 × 65 min ≈ 23 FTE-years at 2,000 h

≈ 4.1 bn UZS/yr

23 FTE × 15M/month × 12 — officer time only; at a 10% addressable share ≈ 1.4 bn

Payback: the ≈ 190M pilot itself (5 of 181 branches, 3 months) does not pay back — it measures the two assumptions. After roll-out the saving of ≈ 4.1 bn/yr repays the pilot within its first month; at a 10% addressable share, within two months. Not counted: faster time-to-decision for the borrower, fewer ПДН compliance findings, better risk selection — claimed only after the pilot measures them.

Who pays: MKBANK (credit department / digital budget). Who owns: micro & family-business credit department; IT deploys; mahalla agents and loan officers use it.

Hard to copy in a weekend — and what comes next

The replication test — what the same LLM tools do not give you

- Statement parsers that learn columns from the data rows (two Uzbek layouts today, any text-layer PDF, XLSX, CSV) and reconcile the balance chain
- A categoriser lexicon in Uzbek Latin, Cyrillic and Russian with counterparty-pattern rules — and the pilot plan to grow it on bank data
- A rule engine mapped clause by clause to the live ПДН regulation, with its exemptions and the back-solved maximum amount
- A memo verifier that makes unsourced numbers impossible — zero hallucination by construction, no LLM needed
- A declared, calibrated synthetic population with rendered documents, a camera model and labelled tampering — a test bed the bank can keep using
- Evaluation through the real app, reproducible with one command each

Next steps

- Week 1–2: scanned-statement OCR page by page; lexicon growth on anonymised bank descriptions; Uzbek/Russian memo templates
- Month 1: shadow mode in 5 branches; core-banking pull for MKBANK's own account holders; credit-bureau REST for obligations
- Month 2–3: assisted mode; retrain the risk model on bank history inside the perimeter; set risk bands with the bank's risk appetite; fairness review by region
- After the pilot: Central Bank conversation — the same engine computes ПДН uniformly for any lender, a supervisory check at zero marginal cost
- Product scope today: Ommabop 2, Express, Business Online 2 (ПДН applies) and family-programme loans (exempt from the cap, capacity still assessed)

Daromad

Income evidence for borrowers who have no payslip — regulation-native, perimeter-native, verified.

Live: https://tys-fintech.vercel.app
Repo: https://github.com/hmunisbekk-droid/TYS-fintech
Team TYS

Appendix A · ПДН rule engine, clause by clause

Clause	Rule as implemented	Where it shows
cl.1	ПДН = average monthly payments on loans/microloans ÷ average monthly income (borrower + co-borrower)	formula panel
cl.2	Includes the application under review and all loan agreements with all lenders; co-borrower loans counted. Lender-type reliefs (2024): banks («банк томонидан…») — only a portfolio quota, up to 15% of the number of loans to individuals outside the requirement; MFOs and pawnshops («микромолия ташкилоти ва ломбард томонидан…») — excluded when amount + outstanding principal ≤ 50 BRV; MFOs lending to the self-employed — 75 BRV. Daromad defaults to bank: ratio always computed, quota shown as a note	lender-type selector, ПДН panel note
cl.4	Guarantor loans count only when payments are more than 30 days overdue («30 кундан ортиқ муддати ўтган тўловлари мавжуд бўлса»)	obligations table, DPD column
cl.4[1]	Terms over 36 months: total payments spread over 36 («барча тўловлар 36 ойга тенг миқдорда тақсимланади»); mortgages over 180 months spread over 180	per-line note
cl.6	Calculation date within 10 calendar days of the loan date	case date
cl.7	Average monthly income = arithmetic mean of the last 12 months; co-borrower income added	income base
cl.9	Evidence: employer salary data, pension, funded-pension contributions, income credited to bank accounts, taxes paid, interest/dividends/rent; other data allowed	memo section 2
cl.10	Ratio ≤ 50% from 1 January 2025 (60% between 1 July 2024 and 1 January 2025)	cap, pass/fail
cl.11	Family-entrepreneurship programme loans exempt from the ratio requirement	exemption badge

BRV = 440,000 UZS from 1 September 2026 (previously 412,000); 50 BRV = 22,000,000 UZS. Clause mapping checked against lex.uz/docs/4654149 and 6809992 on 10 Oct 2026; quotes verbatim in docs/REGULATION_QUOTES.md.

Appendix B · Failure cases, perimeter, human in the loop

Known failure cases (also on the Model card page)

- Scanned statements without a text layer: not OCR'd page by page yet — ask for the XLSX/CSV export
- One signed amount column and no balance column: signs taken from keywords only
- Merchant strings outside the lexicon (mixed scripts, transliteration) → default category, visible in the table
- Dark or heavily skewed receipt photos: OCR confidence < 60% → fields flagged for confirmation, never guessed
- Cash-only businesses: nothing to reconstruct — the memo says so; the officer attests
- Synthetic-to-real gap of the risk score until retrained inside the bank; risk bands are placeholders

Perimeter, audit, human in the loop

- Prototype: static site; documents are parsed in the browser and never transmitted; nothing is stored server-side
- Production: same bundle on the bank's web server (Docker, no outbound network), SSO identity, audit log and cases in the bank DB
- Own customers' statements pulled from core banking; uploads only for other banks' statements and wallets
- Every analysis stamped with rule-engine, model and categoriser versions; hash-chained SHA-256 log; export with the case
- The engine recommends; the officer decides; overrides require a reason; committee above branch limits
- Personal data processed inside the bank with recorded consent; retention per bank policy; no gender or activity-type features in the model

Appendix C · Data sources and licences

Source	Used for	Licence / terms
Daromad synthetic data set (synth/, seed 42)	training, evaluation, demo samples	MIT, this repo; generator and parameters declared; no real person
Berka / PKDD'99 Financial data set (mirror github.com/jlacko/berka-dataset)	offline methodology check	academic challenge data, no licence stated; evaluation only, not redistributed
Tesseract tessdata_fast (uzb, uzb_cyrl, rus, eng); tesseract.js	receipt OCR	Apache 2.0
pdf.js; SheetJS; Next.js; React	parsing, application	Apache 2.0 / MIT
XGBoost; scikit-learn; NumPy; pandas; OpenCV; Playwright; Jinja2; qrcode	training, rendering, evaluation	Apache 2.0 / BSD-3 / MIT
CBU regulation texts (lex.uz 4654149, 6809992)	rule engine, citations	public legal information
stat.uz wages; CBU statistics; MKBANK public pages (sponsored article, bank's own figures)	calibration anchors, business case	public information, cited

Rules compliance: no hard-coded answers (every figure computed from uploaded documents at run time); no real personal or confidential data; all code written during the final by team TYS with LLM coding assistance (declared); deployment from the submitted repository; CI on every push.